

Cloud and Network Security.

Azure Firewall (Microsoft Azure).

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Introduction.

This lab focuses on strengthening network security by deploying and configuring **Azure Firewall** to manage inbound and outbound traffic. The objective is to build a secure virtual environment with defined access rules. This includes creating a virtual network, deploying virtual machines, and setting up routing, firewall rules, and DNS configurations to enforce strict traffic control.

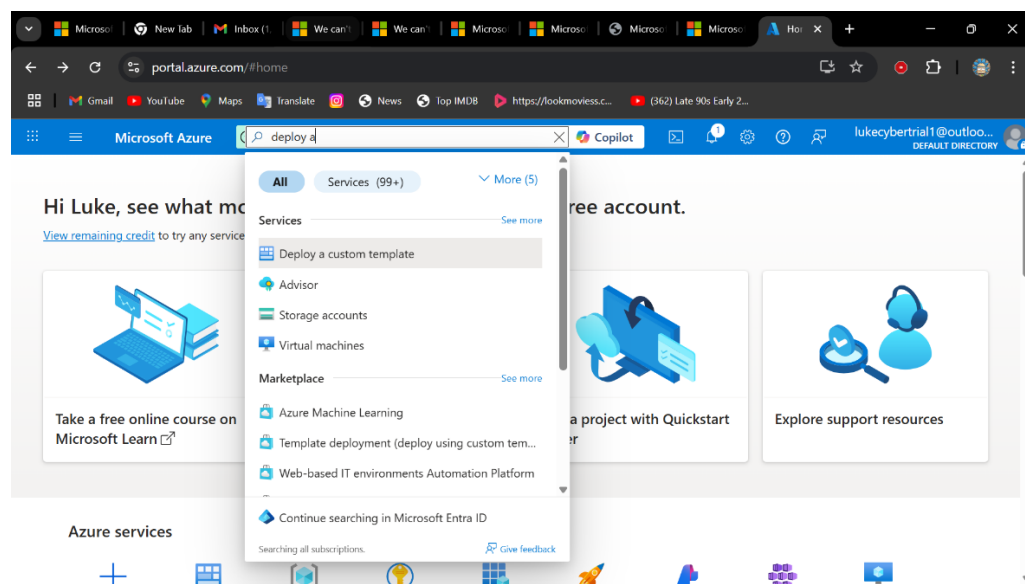
All resources deploy to the **East US** region, following a structured process that simulates a real-world firewall implementation for a workload environment. The lab includes seven core tasks, starting from template-based deployment to firewall rule validation through testing.

Azure Firewall Task

Use a Template to Deploy the Lab Environment

To begin the lab setup, I signed in to the Azure portal using an account with the necessary permissions. From the search bar, I looked up **“Deploy a custom template”** and selected the custom deployment option. I then chose **“Build your own template in the editor”**, clicked on **Load file**, and uploaded the provided ARM template from the path [template.json]. This template was configured to deploy a virtual machine running **Windows Server 2016 Datacenter**.

After reviewing the contents of the template, I clicked **Save** to proceed. The deployment was successfully initiated, laying the foundation for the subsequent firewall and network configurations.



On the **Custom deployment** blade, I configured the required settings for the lab. I selected the appropriate subscription and created a new resource group named **AZ500LAB08**. For the location, I set it to **East US**, as specified in the lab instructions. I then entered a secure password of my choice for the virtual machines, making sure it met Azure's complexity requirements since this password would be needed later to access the VMs.

The screenshot shows the 'Custom deployment' blade in the Azure portal. The 'Project details' section includes a 'Subscription' dropdown set to 'Azure subscription 1' and a 'Resource group' dropdown set to '(New) AZ500LAB08'. The 'Instance details' section includes a 'Region' dropdown set to 'East US', an 'Admin Username' field set to 'localadmin', and an 'Admin Password' field with masked characters. Below these are several input fields for virtual machines and networks, all with checkmarks indicating they are configured. At the bottom, there are buttons for 'Previous', 'Next', and 'Review + create'.

Once all settings were in place, I clicked **Review + create**, reviewed the configuration summary, and proceeded to click **Create** to start the deployment. The deployment began successfully and took approximately 2 minutes to complete.

The screenshot shows the 'Microsoft.Template-20250714180539 | Overview' blade in the Azure portal. The 'Overview' tab is selected, showing a green checkmark and the text 'Your deployment is complete'. Below this, the 'Deployment details' section shows the deployment name, subscription, and resource group. The 'Next steps' section includes a 'Go to resource group' button. On the right side, there are several informational cards for 'Cost management', 'Microsoft Defender for Cloud', 'Free Microsoft tutorials', and 'Work with an expert'. The bottom of the screen shows the Windows taskbar with the time 6:06 PM on 7/14/2025.

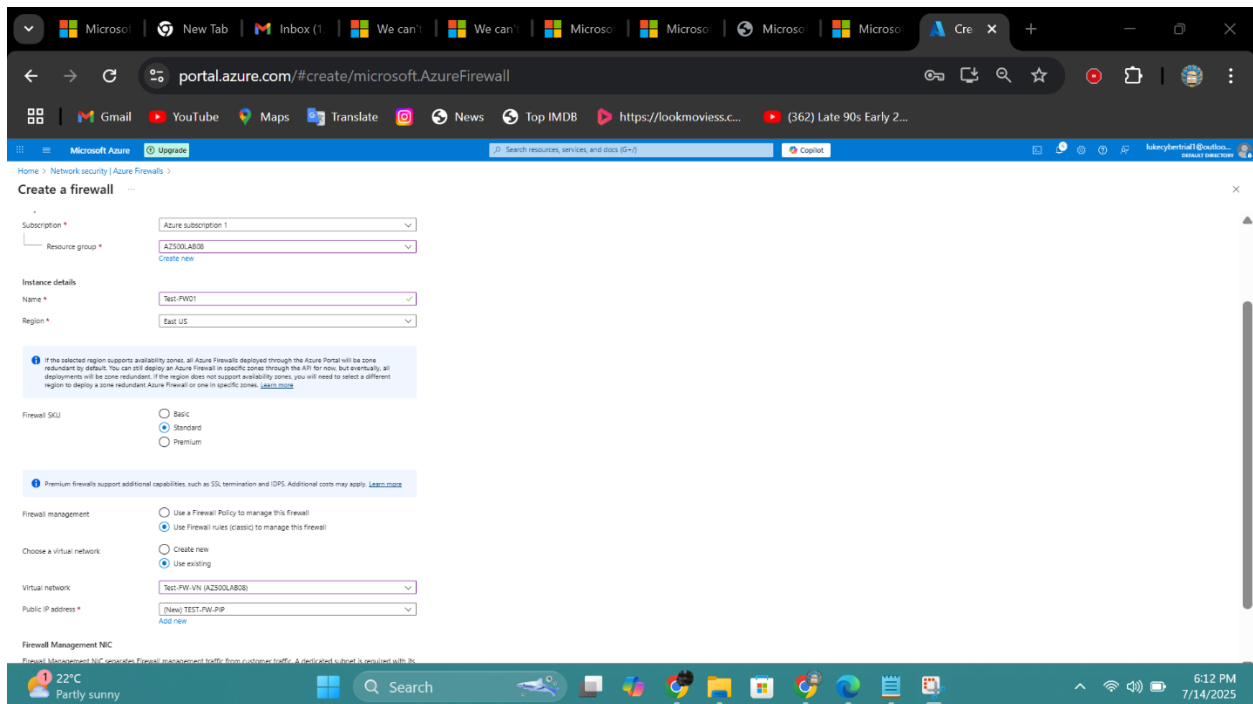
Deploy the Azure Firewall

To enhance the security posture of the environment, I deployed an Azure Firewall into the virtual network.

I started by searching for “**Firewalls**” in the Azure portal and selected the **+ Create** option. On the **Basics** tab of the **Create a firewall** blade, I configured the following:

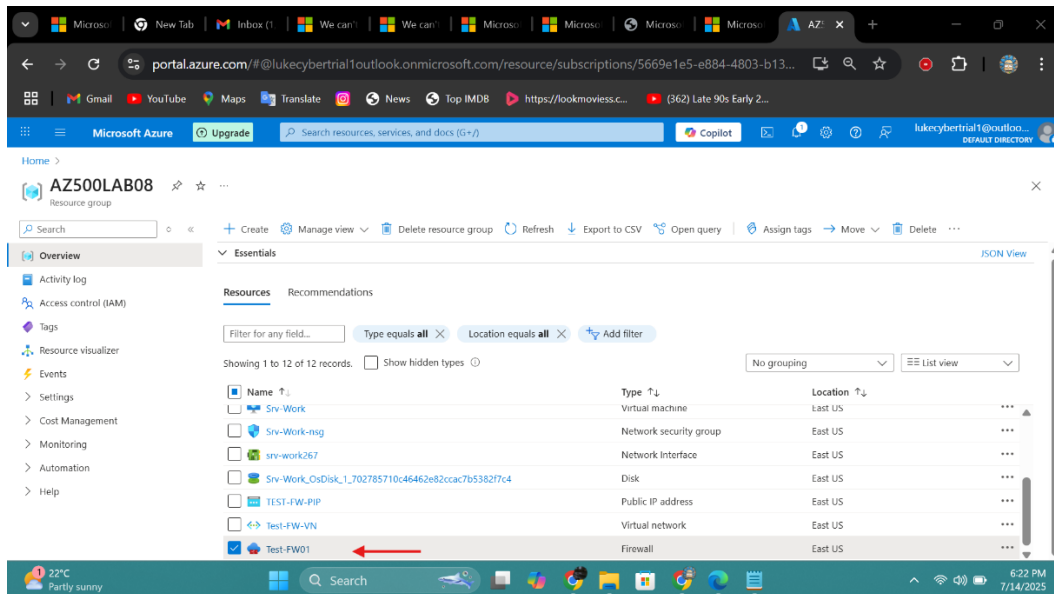
- **Resource group:** AZ500LAB08
- **Firewall name:** Test-FW01
- **Region:** East US
- **Firewall SKU:** Standard
- **Firewall management:** Use Firewall rules (classic)
- **Virtual network:** Used existing Test-FW-VN
- **Firewall Management NIC:** Disabled by unchecking the option
- **Public IP:** Added a new IP named TEST-FW-PIP and confirmed by clicking OK

After completing the setup, I clicked **Review + create**, validated the settings, and then clicked **Create** to deploy the firewall.

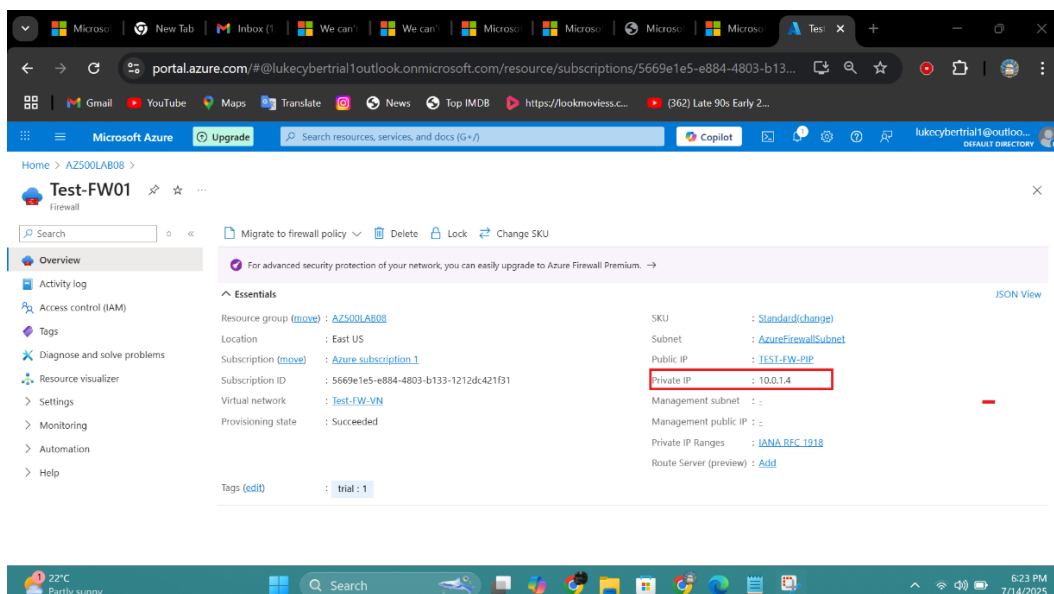


After the firewall deployment completed, I returned to the Azure portal and used the search bar to look for “**Resource groups**”. From the list, I selected the resource group named **AZ500LAB08**.

On the **AZ500LAB08** resource group blade, I reviewed the list of deployed resources. To make navigation easier, I sorted the list by **Type**. Among the listed items, I located and clicked on the entry representing the **Test-FW01** firewall to begin reviewing and configuring its settings.



On the **Test-FW01** firewall blade, I reviewed the firewall's overview section to identify the **Private IP address** assigned during deployment. This IP is automatically allocated from the subnet range of the virtual network and is used for internal routing and firewall rule enforcement. The private IP address was clearly displayed under the **Private IP address** field in the **Firewall Management** section of the blade.

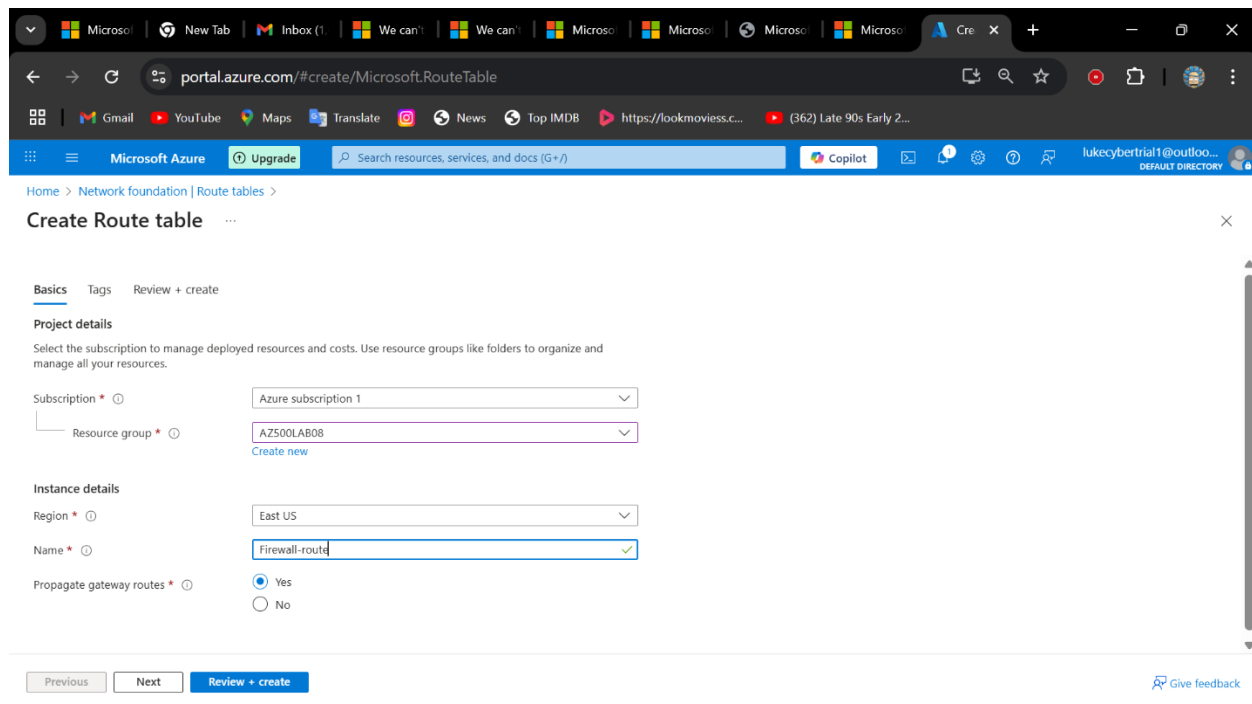


Create a Default Route

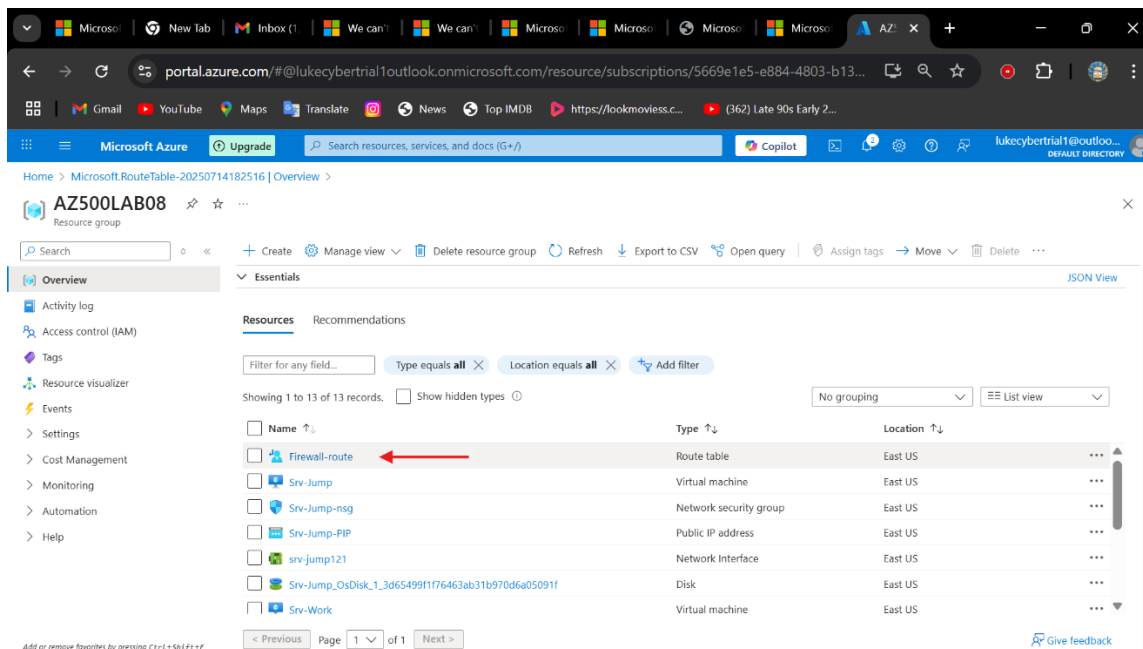
To ensure that outbound traffic from the workload subnet is routed through the firewall, I created a custom route table. I began by searching for “**Route tables**” in the Azure portal and selected the **+ Create** option. On the **Create route table** blade, I configured the following settings:

- **Resource group:** AZ500LAB08
- **Region:** East US
- **Name:** Firewall-route

After confirming the settings, I clicked **Review + create**, then **Create** to deploy the route table. I waited for the provisioning process to complete before proceeding to associate the route with the subnet.

The screenshot shows the 'Create Route table' blade in the Azure portal. The browser address bar shows 'portal.azure.com/#create/Microsoft.RouteTable'. The page has a blue header with the Microsoft Azure logo, a search bar, and a 'Copilot' button. The breadcrumb trail is 'Home > Network foundation | Route tables >'. The main heading is 'Create Route table'. Below this are tabs for 'Basics', 'Tags', and 'Review + create'. The 'Basics' tab is active. Under 'Project details', there is a note about selecting a subscription and resource group. The 'Subscription' dropdown is set to 'Azure subscription 1' and the 'Resource group' dropdown is set to 'AZ500LAB08'. Under 'Instance details', the 'Region' dropdown is set to 'East US' and the 'Name' text box contains 'Firewall-route'. There is a 'Create new' link below the resource group dropdown. The 'Propagate gateway routes' section has a radio button selected for 'Yes'. At the bottom, there are three buttons: 'Previous', 'Next', and 'Review + create'. A 'Give feedback' link is in the bottom right corner.

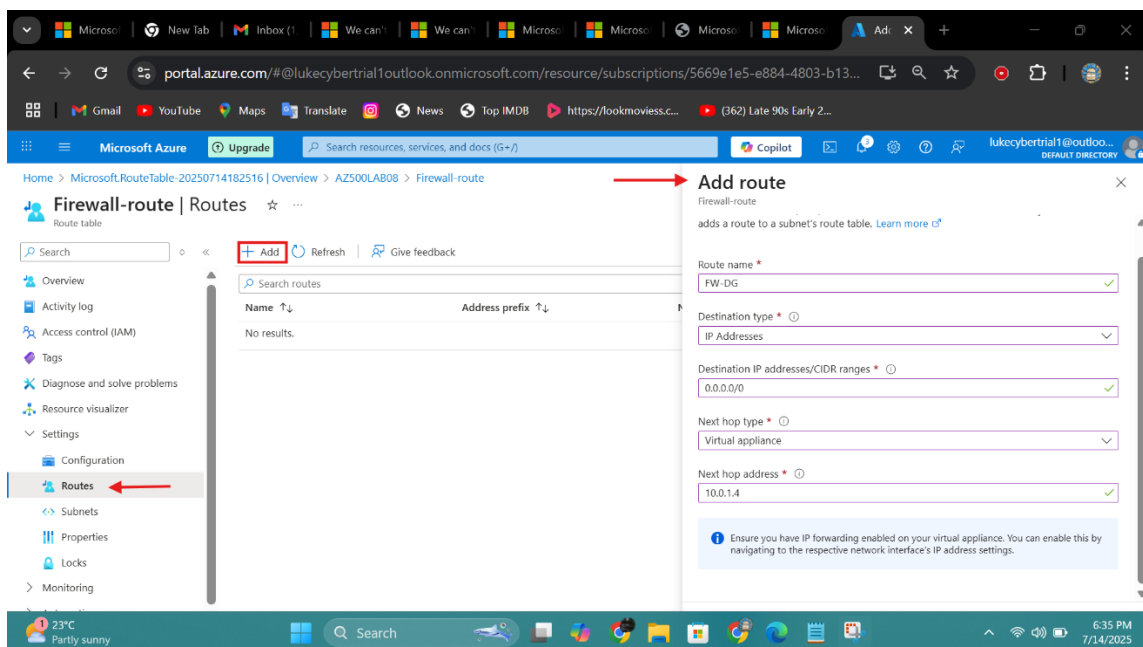
Once the route table deployment completed, I returned to the **Route tables** blade and clicked **Refresh** to update the list. From the refreshed list, I located and clicked on the entry named **Firewall-route** to open its configuration and prepare it for custom route settings.



On the **Firewall-route** blade, I navigated to the **Settings** section and selected **Subnets**. From the **Firewall-route | Subnets** blade, I clicked on **+ Associate** to bind the route table to the appropriate subnet. In the **Associate subnet** blade, I set the following:

- **Virtual network:** Test-FW-VN
- **Subnet:** Workload-SN

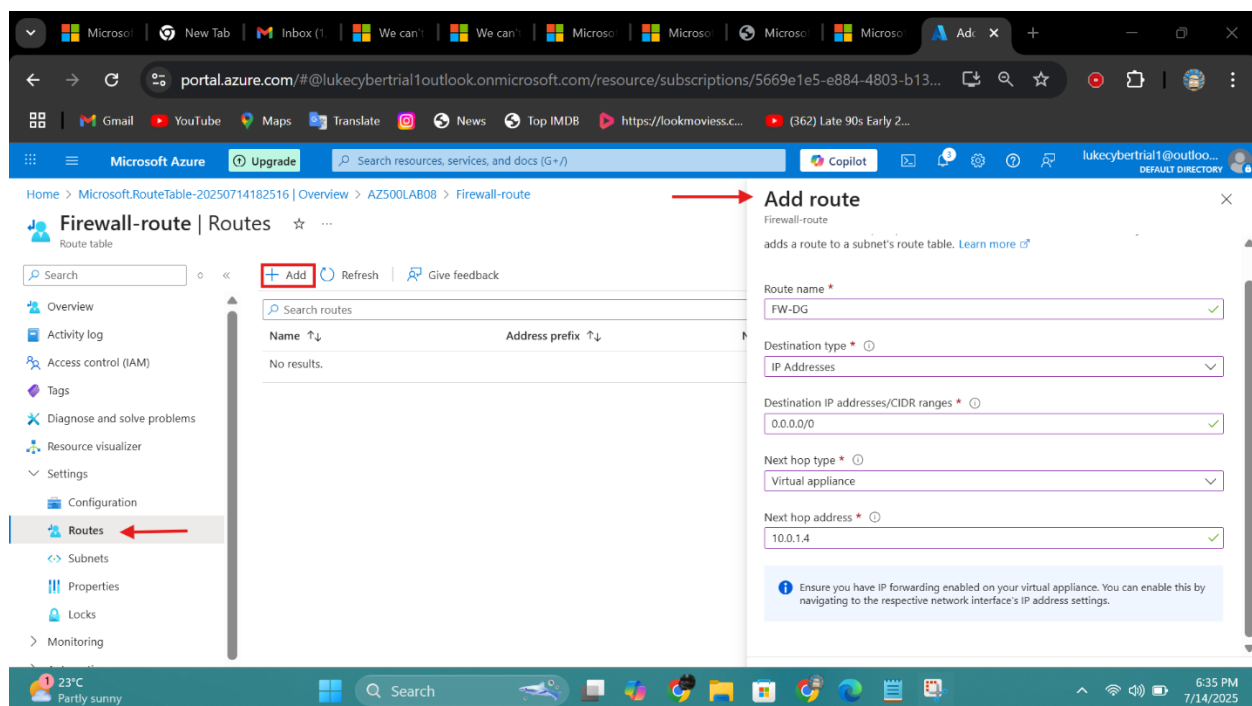
I ensured that **Workload-SN** was selected, as this subnet is meant to have all its outbound traffic routed through the firewall. After verifying the settings, I clicked **OK** to complete.



Back on the **Firewall-route** blade, I went to the **Settings** section and selected **Routes**, then clicked **+ Add** to define a custom route. On the **Add route** blade, I entered the following configuration:

- **Route name:** FW-DG
- **Destination Type:** IP Address
- **Destination IP address/CIDR ranges:** 0.0.0.0/0 (to match all outbound traffic)
- **Next hop type:** Virtual appliance
- **Next hop address:** The **private IP address** of the **Test-FW01** firewall, which I had identified earlier

Although Azure Firewall is a managed service, selecting **Virtual appliance** as the next hop type is appropriate in this context for routing purposes. I finalized the setup by clicking **Add**, successfully applying the route to the table.



Configure an Application Rule

To control outbound access from the workload subnet, I configured an application rule that specifically allows access to www.bing.com. I navigated back to the **Test-FW01** firewall in the Azure portal. From the **Settings** section, I selected **Rules (classic)**. Within the **Rules (classic)** blade, I switched to the **Application rule collection** tab and clicked **+ Add application rule collection**. On the configuration blade, I set the following:

- **Name:** App-Coll01
- **Priority:** 200
- **Action:** Allow

Under the **Target FQDNs** section, I added a new entry with the following details:

- **Name:** AllowGH
- **Source type:** IP Address
- **Source:** 10.0.2.0/24 (the workload subnet address range)
- **Protocol port:** http:80, https:443
- **Target FQDNs:** www.bing.com

Once all values were entered and verified, I clicked **Add** to create the rule. This rule ensures that only traffic from the workload subnet can access www.bing.com, using standard web ports.

Task 5: Configure a Network Rule

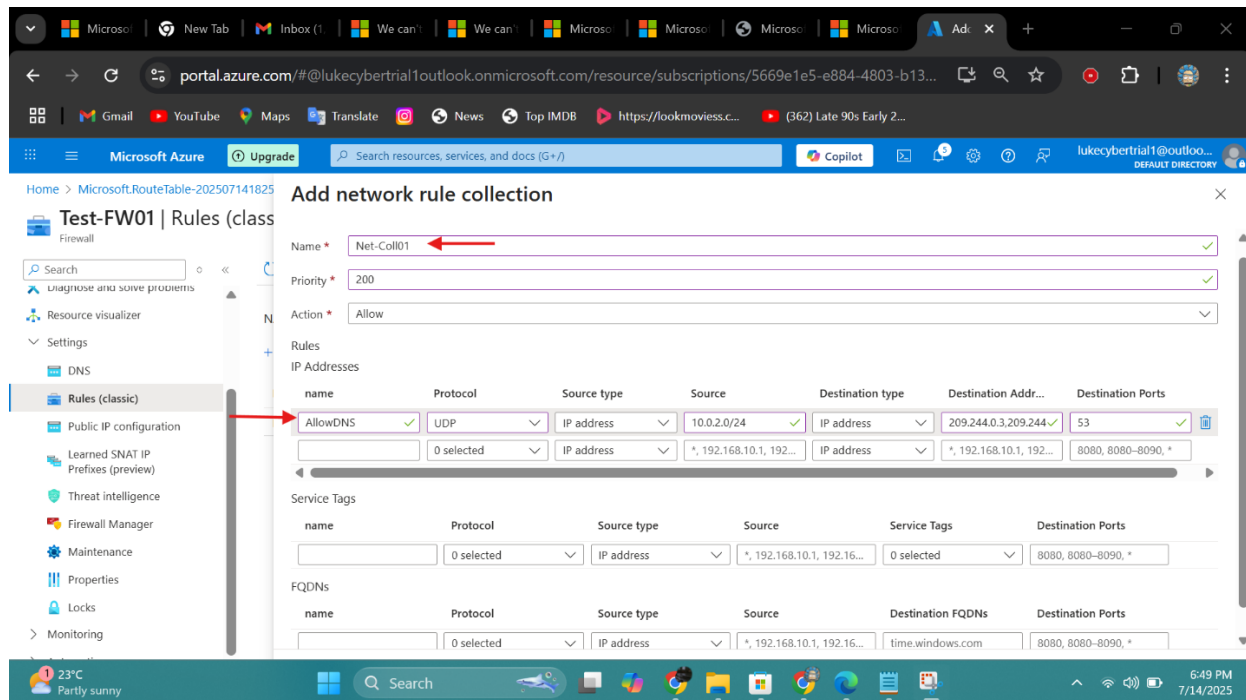
To allow DNS resolution from the workload subnet, I created a network rule that permits outbound traffic to specific public DNS servers on UDP port 53. I returned to the **Test-FW01 | Rules (classic)** blade in the Azure portal. From there, I navigated to the **Network rule collection** tab and clicked **+ Add network rule collection**. In the **Add network rule collection** blade, I configured the following:

- **Name:** Net-Coll01
- **Priority:** 200
- **Action:** Allow

Within the **IP Addresses** section, I added a rule with these details:

- **Name:** AllowDNS
- **Protocol:** UDP
- **Source type:** IP address
- **Source address:** 10.0.2.0/24 (representing the workload subnet)
- **Destination type:** IP address
- **Destination addresses:** 209.244.0.3, 209.244.0.4 (public DNS servers)
- **Destination ports:** 53

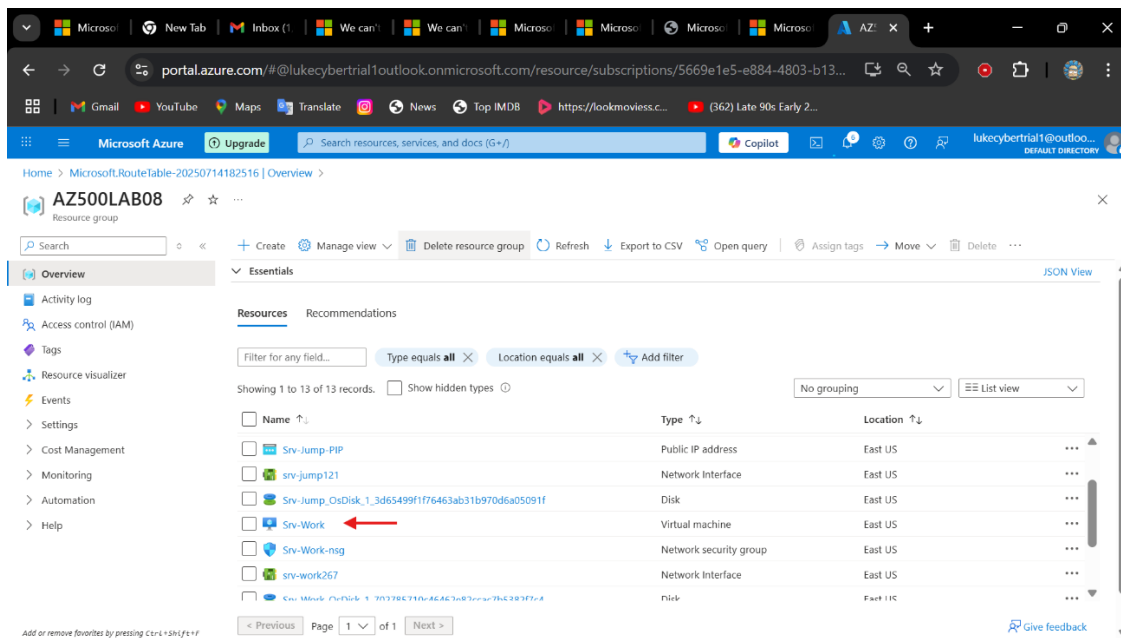
Once all fields were completed, I clicked **Add** to finalize the network rule. This configuration ensures that only traffic from the workload subnet can perform DNS lookups via those two external DNS servers.



Configure the Virtual Machine DNS Servers

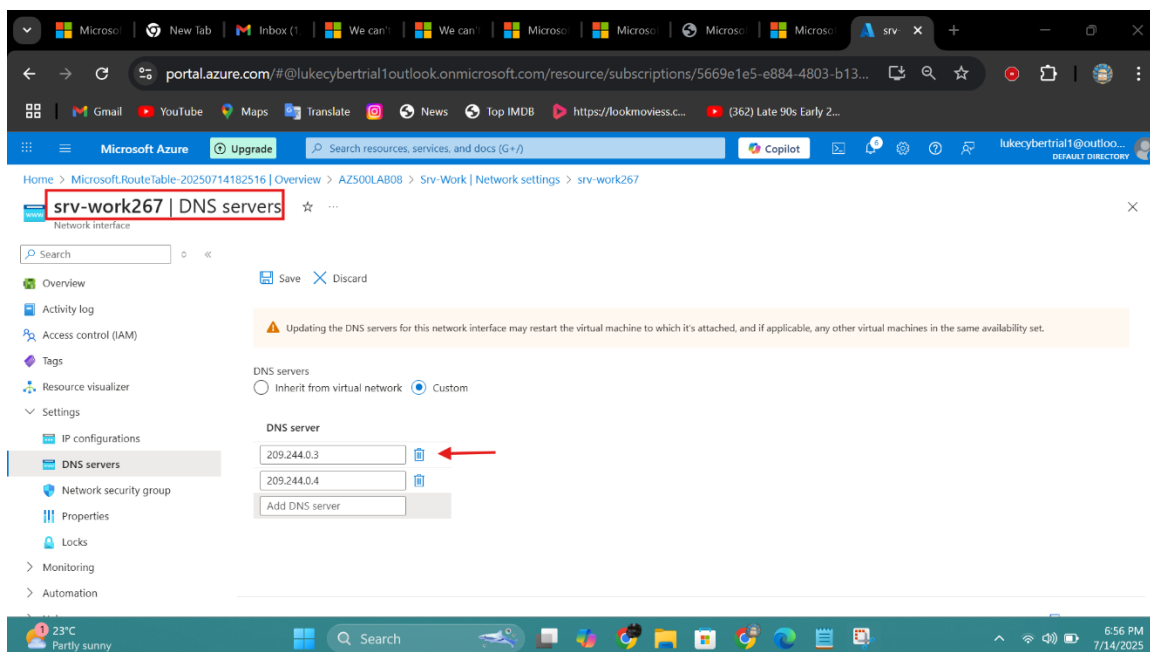
Although not directly related to the firewall configuration, I proceeded to set up custom DNS servers for the virtual machine to ensure it resolves domain names through the intended external servers.

From the Azure portal, I navigated back to the **AZ500LAB08** resource group. On the **AZ500LAB08** blade, I reviewed the list of deployed resources and located the virtual machine named **Srv-Work**. I then selected it to begin configuring its DNS settings.



From the **Srv-Work** virtual machine blade, I selected the **Networking** option in the left-hand menu. On the **Srv-Work | Networking** blade, I clicked the link under the **Network interface** entry to open its configuration. Within the **network interface** blade, I navigated to the **DNS servers** section under **Settings**. I selected the **Custom** option, then entered the two DNS server IPs used earlier in the network rule:

- 209.244.0.3
- 209.244.0.4



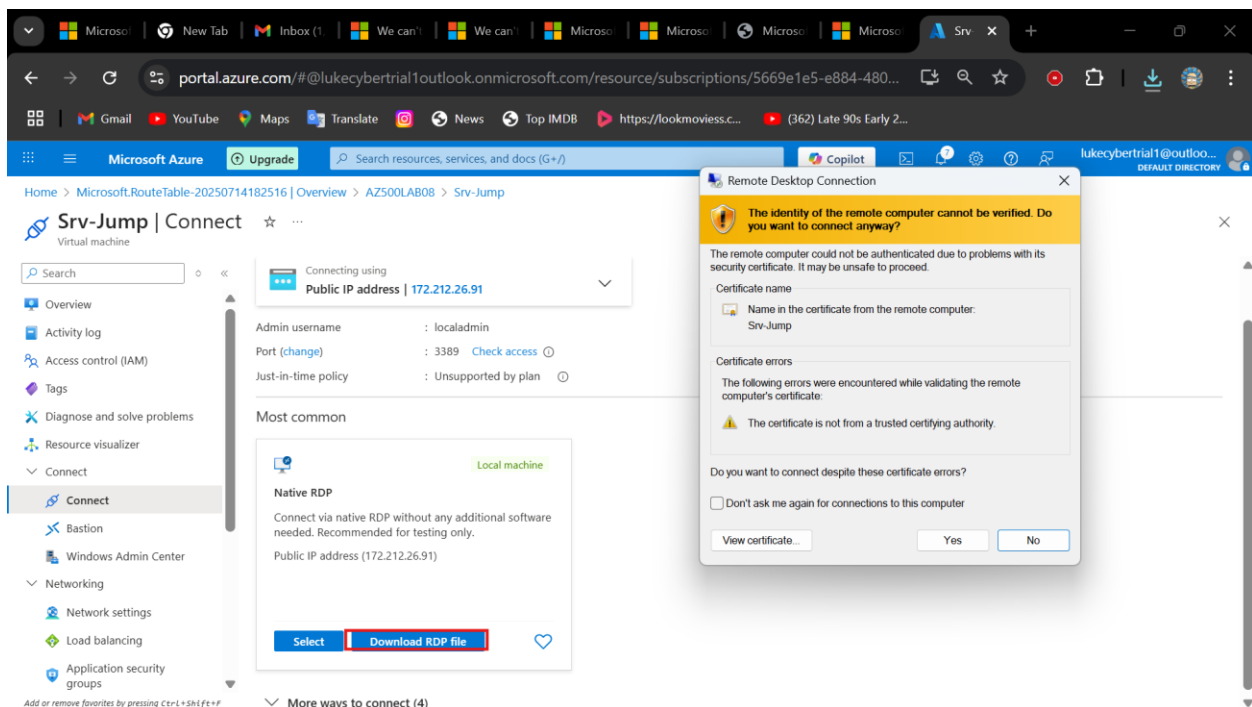
After entering both addresses, I clicked **Save** to apply the change. Upon saving, Azure automatically initiated a restart of the **Srv-Work** virtual machine, as updating the DNS servers at the NIC level requires a reboot. I waited for the update and restart to complete before proceeding.

Test the Firewall

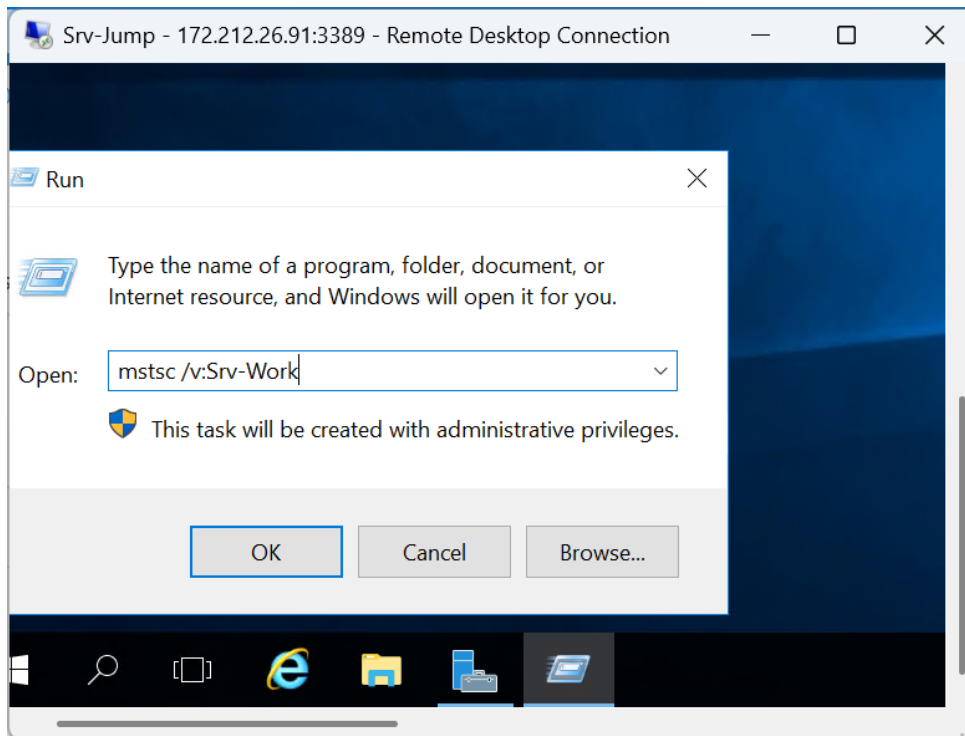
To confirm the firewall is correctly enforcing outbound rules, I initiated a test from within the lab environment. I began by returning to the **AZ500LAB08** resource group in the Azure portal. From the list of resources, I selected the **Srv-Jump** virtual machine.

On the **Srv-Jump** blade, I clicked **Connect**, then selected **RDP** from the drop-down menu. I downloaded the RDP file and launched a Remote Desktop session to the VM. When prompted for credentials, I used the following:

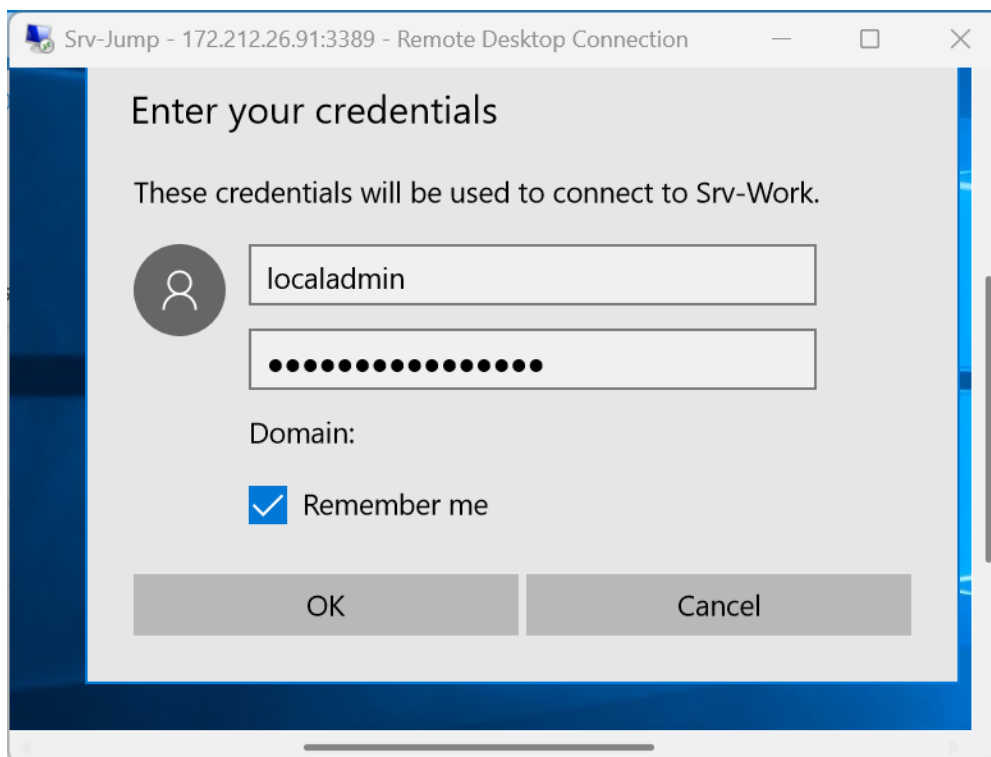
- **Username:** localadmin
- **Password:** the secure password configured during Task 1



While inside the **Remote Desktop** session to the **Srv-Jump** virtual machine, I initiated a connection to the workload VM (**Srv-Work**) to simulate user activity from within the secured subnet. I right-clicked the **Start** menu and selected **Run**. In the **Run dialog box**, I entered the following command to launch an RDP session to Srv-Work **[mstsc /v:Srv-Work]**.

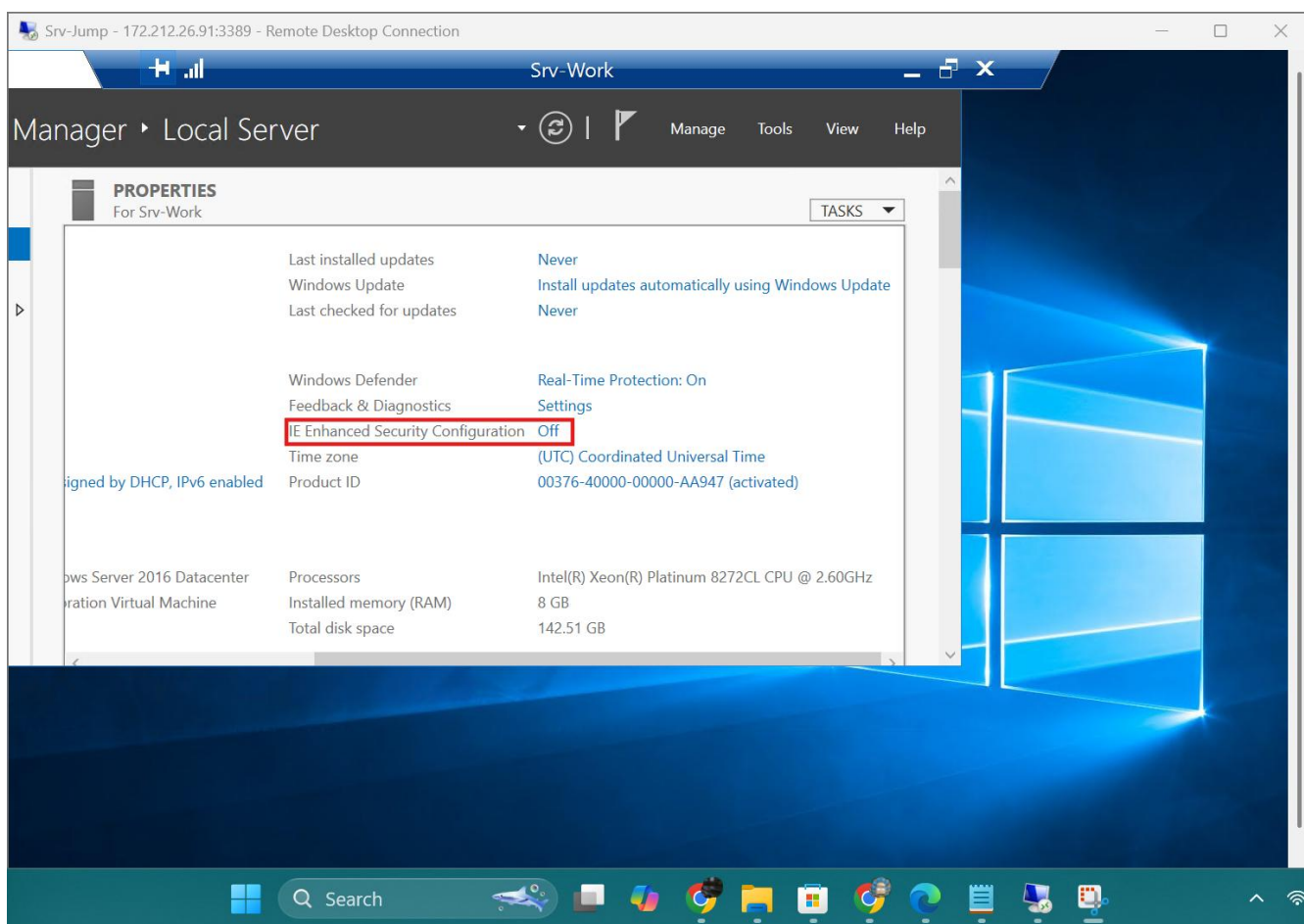


This command triggered the Remote Desktop Connection window, targeting the **Srv-Work** VM by name. I proceeded to authenticate using the same credentials provided earlier during deployment. This step was necessary in order to simulate browsing activity from inside the **Workload-SN** subnet and validate that traffic was being filtered correctly by **Azure Firewall**.

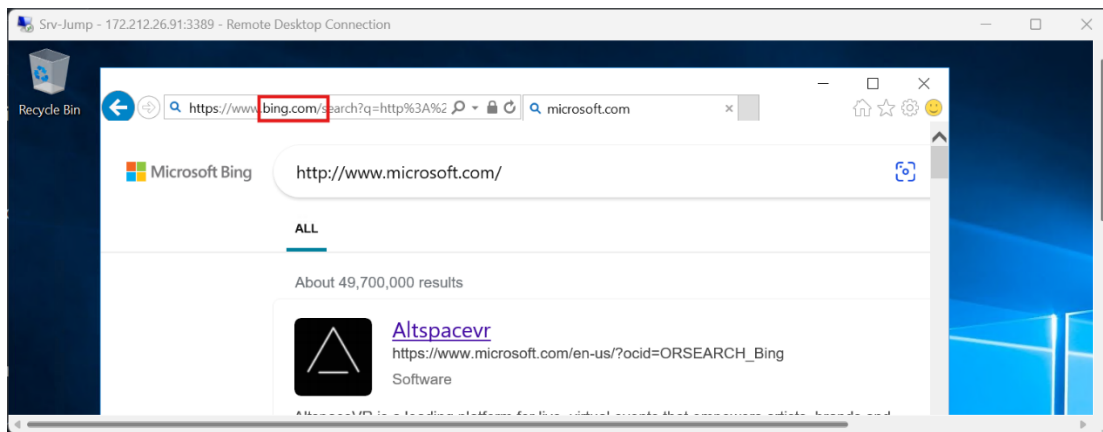


Once connected to the **Srv-Work** virtual machine via Remote Desktop, I opened **Server Manager**. From the left-hand menu, I clicked on **Local Server** to access the system's configuration settings. In the **Properties** pane, I located the setting labeled **IE Enhanced Security Configuration** and clicked on the current status link.

In the **Internet Explorer Enhanced Security Configuration** dialog box, I set both options — for **Administrators** and **Users** — to **Off**, then clicked **OK** to apply the changes. This adjustment was made to ensure that Internet Explorer would not block or restrict access to external websites due to strict security prompts, which could interfere with the firewall test.



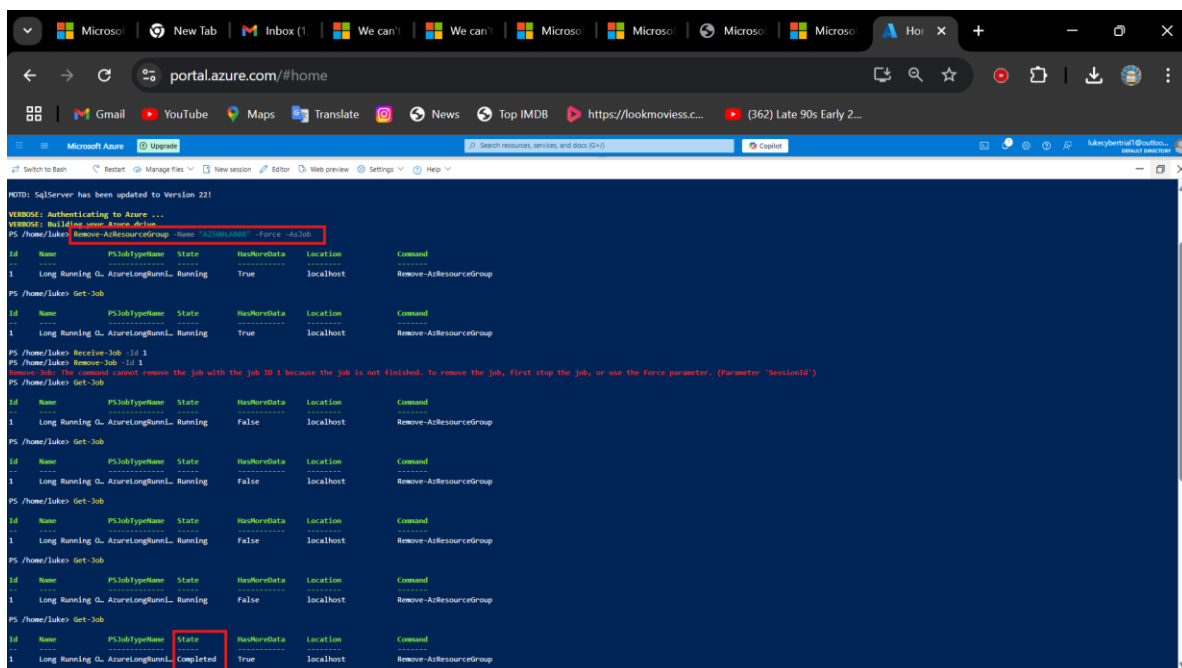
With Internet Explorer now unrestricted on the **Srv-Work** virtual machine, I launched the browser and navigated to <https://www.bing.com>. The website loaded successfully, confirming that the **Azure Firewall application rule** was working as intended. Traffic from the **10.0.2.0/24** subnet (which includes Srv-Work) was allowed to reach www.bing.com over ports **80** and **443**, as configured in the earlier application rule. This successful connection validated that the firewall was correctly inspecting and permitting outbound HTTP/HTTPS traffic only to the specified FQDN.



Session Termination and Resource Clean-Up

After successfully confirming that the firewall rule permitted access to www.bing.com, I proceeded to close both **Remote Desktop sessions** — one to **Srv-Jump** and the other to **Srv-Work** — completing the firewall validation phase of the lab.

To avoid incurring any unnecessary charges, I initiated a cleanup of all lab-related resources. In the **Azure portal**, I launched the **Cloud Shell** by clicking the shell icon at the top-right of the interface. When prompted, I selected **PowerShell** and allowed the creation of a storage account if required. With **PowerShell** active in the Cloud Shell, I executed the following command to delete the resource group and all associated resources: **[Remove-AzResourceGroup -Name "AZ500LAB08" -Force -AsJob]**. This command initiated an asynchronous job that began tearing down all infrastructure deployed during the lab, including the virtual network, firewall, route tables, and virtual machines.



Summary

In this lab, I deployed and tested **Azure Firewall** using the **Azure portal** and **Cloud Shell (PowerShell)** to control outbound traffic within a virtual network.

Tools Used:

- Azure Portal
- Azure Resource Manager (ARM) template
- Cloud Shell with PowerShell
- Remote Desktop (RDP)
- Internet Explorer

Resources Created:

- **Virtual Network** with two subnets:
 - Workload-SN (for the internal VM)
 - Jump-SN (for the jump host)
- **Virtual Machines:**
 - Srv-Work
 - Srv-Jump
- **Azure Firewall** with:
 - Public IP (TEST-FW-PIP)
 - Custom route table (Firewall-route)

Configuration Tasks:

- Deployed the environment using a custom ARM template
- Configured Azure Firewall with:
 - **Application rule** allowing access to www.bing.com
 - **Network rule** permitting DNS queries to 209.244.0.3 and 209.244.0.4 via UDP port 53
- Manually set custom DNS servers on the Srv-Work NIC

- Verified rule enforcement by accessing Bing in Internet Explorer from Srv-Work

Cleanup:

Deleted all resources using the following command in Cloud Shell: ***[Remove-AzResourceGroup -Name "AZ500LAB08" -Force -AsJob]***

Conclusion

This lab demonstrated the practical deployment and configuration of Azure Firewall to manage and secure outbound traffic in a virtual network. By implementing custom routing, application rules, and network rules, I was able to enforce precise access controls over VM traffic. The successful test of allowed web and DNS access validated the firewall's effectiveness. Overall, the lab reinforced key concepts in network security, traffic filtering, and resource management within Azure.